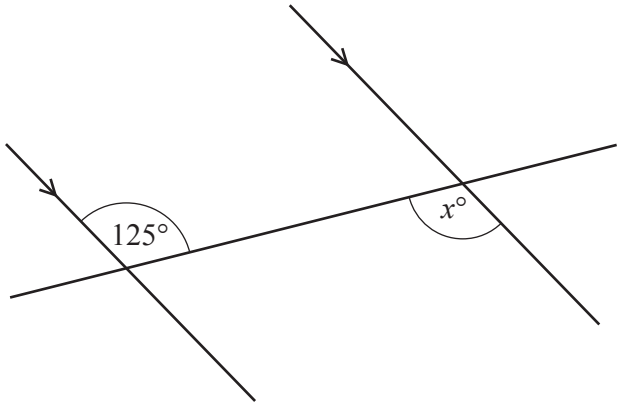
5 **(a)**



NOT TO SCALE

The diagram shows a pair of parallel lines and a straight line.

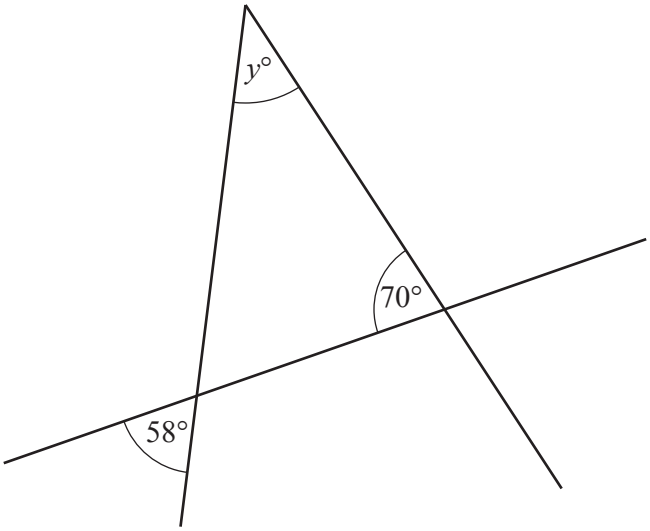
(i) Write down the mathematical name for the type of angle marked 125°.

..... [1]

(ii) Give the geometrical reason why the value of x is 125.

..... [1]

(b)



NOT TO SCALE

The diagram shows three straight lines.

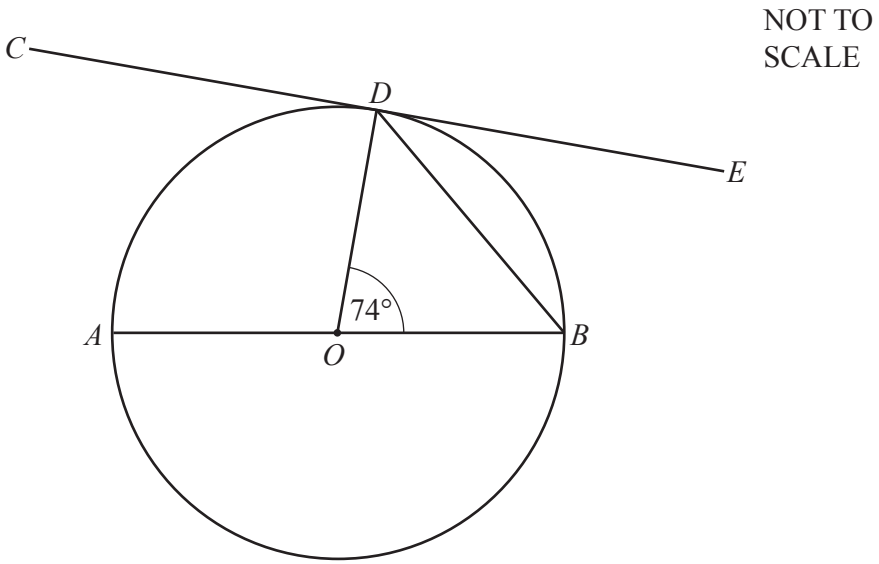
Find the value of y.
Write down the geometrical properties needed to find the value of y.

.....

.....

$y =$ ..... [3]

(c)



The diagram shows a circle, centre O, with diameter AOB.
The line CDE touches the circle at D and angle $DOB = 74°$.

(i) Write down the mathematical name of the line CDE.

..... [1]

(ii) Work out angle ODB.

Angle $ODB =$ ..... [2]

(iii) Work out angle BDE.
Give a geometrical reason for your answer.

Angle $BDE =$ ..... because .....

..... [2]

(d) Find the interior angle of a regular 15-sided polygon.

..... [2]